BrucellaID

Brucella species identification and MLST from reads

Quality assurance report

4 sample(s): 4 ok, 0 with a warning, 0 failed. BrucellaID 1.0.0, run on 2026-10-31T21:21:05-04:00.

This is an example report: a real run of the four public read sets of the BrucellaID tutorial, produced by scripts/make_example_report.py, which leaves out the computer it ran on. The user account, the host name, the operating system and the working directory read [removed], and the input and reference files are named without the folder they were in. Everything else is as BrucellaID wrote it. A report of your own names them all, which is what makes it worth signing.

Summary

Run started	2026-10-31T21:21:05-04:00
Run finished	2026-10-31T21:21:37-04:00
Operator	M-O Duceppe
Command	python3 scripts/make_example_report.py reads

Brucella abortus — ST 1 — 1 sample(s)

Sample	Species	Biovar	From	Decision	ST	Allele profile	Status
ERR3454415	Brucella abortus	1	MASH+ST	agree	1	2-1-1-2-1-3-1-1-1	ok

Brucella canis — ST 20 — 1 sample(s)

Sample	Species	Biovar	From	Decision	ST	Allele profile	Status
ERR2136546	Brucella canis		MASH+ST	arbitrated	20	1-6-4-1-5-3-5-4-6	ok

Brucella melitensis — ST 11 — 1 sample(s)

Sample	Species	Biovar	From	Decision	ST	Allele profile	Status
SRR19520329	Brucella melitensis	3	MASH+ST	agree	11	3-2-3-2-1-5-3-2-10	ok

Brucella suis — ST 15 — 1 sample(s)

Sample	Species	Biovar	From	Decision	ST	Allele profile	Status
SRR8550484	Brucella suis	2	MASH+ST	agree	15	1-2-7-1-3-3-5-3-2	ok

Reviewed by

Name Date Signature

Notes, warnings and errors

Sample	Status	What it says
ERR2136546	ok	nearly as close: B-REF-BS4-40.fasta (Brucella suis bv. 4, 768/1000 hashes)
ERR2136546	ok	MASH cannot separate Brucella canis and Brucella suis: ST 20 is Brucella canis in 70 of 70 isolate(s), which agrees with the closest reference

ERR3454415 ok Brucella abortus ST 1

Identification: the MASH distances

Closest reference	NC_006932-NC_006933.fasta
What it is	Brucella abortus bv. 1 str. 9-941
NCBI accessions	NC_006932.1, NC_006933.1
MASH distance	0.004549
Estimated identity	99.545%
Shared min-hashes	833/1000 (at least 150 required)

Reference genome	Species	Biovar	Distance	Identity %	Shared hashes
NC_006932-NC_006933.fasta	Brucella abortus	1	0.004549	99.545	833/1000
NZ_CP007682-NZ_CP007683.fasta	Brucella abortus	3	0.005741	99.426	796/1000
NC_012441-NC_012442.fasta	Brucella melitensis	2	0.006734	99.327	767/1000
NZ_CP007760-NZ_CP007761.fasta	Brucella melitensis	3	0.006769	99.323	766/1000
NZ_CP018508-NZ_CP018509.fasta	Brucella melitensis	1	0.006769	99.323	766/1000
NC_013119_NC_013118.fasta	Brucella microti		0.006946	99.305	761/1000
NC_003317-NC_003318.fasta	Brucella melitensis	1	0.007088	99.291	757/1000
NC_022905-NC_022906.fasta	Brucella ceti		0.007124	99.288	756/1000
GCF_900446125.fasta	Brucella neotomae		0.007304	99.270	751/1000
Bceti1Cudo.fasta	Brucella ceti		0.007486	99.251	746/1000
NC_017251-NC_017250.fasta	Brucella suis	1	0.007523	99.248	745/1000
NC_010103-NC_010104.fasta	Brucella canis		0.007596	99.240	743/1000
B-REF-BS4-40.fasta	Brucella suis	4	0.007633	99.237	742/1000
NC_010169-NC_010167.fasta	Brucella suis	2	0.007781	99.222	738/1000
NZ_CP007719-NZ_CP007718.fasta	Brucella suis	3	0.007893	99.211	735/1000
NC_009505-NC_009504.fasta	Brucella ovis		0.008500	99.150	719/1000

The species call

Species reported	Brucella abortus
Biovar	1 (of 9-941, the closest reference of that species)
Decided by	MASH+ST
Species within reach	Brucella abortus
Sequence type	1
What that ST is	ST 1 is Brucella abortus in 463 of 466 isolate(s) (the rest: Brucella melitensis:3)

agree: The sequence type names the same species as the closest reference genome, which named it on its own: the two kinds of evidence agree.

MLST: the nine allele calls

Sequence type	1
Allele profile	2-1-1-2-1-3-1-1-1
Every locus called	yes
Reads in the input	1,175,082
Reads matching an allele	2,948

Locus	Allele	Reads	Runner-up	Its reads	Ahead	Ratio	Note
gap	2	432	gap_6	300	132	1.44	
aroA	1	322	aroA_30	210	112	1.53	
glk	1	222	glk_21	158	64	1.41	
dnaK	2	340	dnaK_26	222	118	1.53	
gyrB	1	334	gyrB_22	224	110	1.49	
trpE	3	320	trpE_4	242	78	1.32	
cobQ	1	308	cobQ_12	196	112	1.57	
int_hyp	1	368	int_hyp_4	256	112	1.44	
omp25	1	292	omp25_12	186	106	1.57	

Input files

File	Size	Modified	MD5
ERR3454415_1.fastq.gz	87.4 MB	2026-10-31 15:12:48 EDT	8eaddc2c8044657712ec586ba5482c70

File	Size	Modified	MD5
ERR3454415_2.fastq.gz	83.5 MB	2026-10-31 15:13:39 EDT	b2706df339382a63f38bba34222b4053

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ERR3454415_1.fastq.gz
ERR3454415_2.fastq.gz
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ERR2136546 ok Brucella canis ST 20

Identification: the MASH distances

Closest reference	NC_010103-NC_010104.fasta
What it is	Brucella canis str. ATCC 23365
NCBI accessions	NC_010103.1, NC_010104.1
MASH distance	0.006524
Estimated identity	99.348%
Shared min-hashes	773/1000 (at least 150 required)

Reference genome	Species	Biovar	Distance	Identity %	Shared hashes
NC_010103-NC_010104.fasta	Brucella canis		0.006524	99.348	773/1000
B-REF-BS4-40.fasta	Brucella suis	4	0.006698	99.330	768/1000
NC_017251-NC_017250.fasta	Brucella suis	1	0.007196	99.280	754/1000
NZ_CP007719-NZ_CP007718.fasta	Brucella suis	3	0.007268	99.273	752/1000
NC_013119_NC_013118.fasta	Brucella microti		0.008080	99.192	730/1000
NC_010169-NC_010167.fasta	Brucella suis	2	0.008849	99.115	710/1000
Bceti1Cudo.fasta	Brucella ceti		0.008889	99.111	709/1000
GCF_900446125.fasta	Brucella neotomae		0.009086	99.091	704/1000
NC_022905-NC_022906.fasta	Brucella ceti		0.009366	99.063	697/1000
NZ_CP007760-NZ_CP007761.fasta	Brucella melitensis	3	0.009568	99.043	692/1000
NC_006932-NC_006933.fasta	Brucella abortus	1	0.009650	99.035	690/1000
NC_012441-NC_012442.fasta	Brucella melitensis	2	0.009650	99.035	690/1000
NZ_CP007682-NZ_CP007683.fasta	Brucella abortus	3	0.009650	99.035	690/1000
NZ_CP018508-NZ_CP018509.fasta	Brucella melitensis	1	0.009691	99.031	689/1000
NC_003317-NC_003318.fasta	Brucella melitensis	1	0.009938	99.006	683/1000
NC_009505-NC_009504.fasta	Brucella ovis		0.011001	98.900	658/1000

The species call

Species reported	Brucella canis
Decided by	MASH+ST
Species within reach	Brucella canis, Brucella suis
Sequence type	20
What that ST is	ST 20 is Brucella canis in 70 of 70 isolate(s)

arbitrated: A reference of a second species was near enough to the closest one that the distance did not really choose. The sequence type chose between the species it left within reach.

MLST: the nine allele calls

Sequence type	20
Allele profile	1-6-4-1-5-3-5-4-6
Every locus called	yes
Reads in the input	625,576
Reads matching an allele	1,824

Locus	Allele	Reads	Runner-up	Its reads	Ahead	Ratio	Note
gap	1	232	gap_26	170	62	1.36	
aroA	6	194	aroA_31	80	114	2.42	
glk	4	148	glk_2	94	54	1.57	
dnaK	1	244	dnaK_2	162	82	1.51	
gyrB	5	224	gyrB_4	110	114	2.04	
trpE	3	178	trpE_5	114	64	1.56	
cobQ	5	172	cobQ_30	108	64	1.59	
int_hyp	4	234	int_hyp_1	180	54	1.30	
omp25	6	172	omp25_2	98	74	1.76	

Input files

File	Size	Modified	MD5
ERR2136546_1.fastq.gz	57.0 MB	2026-10-31 15:16:00 EDT	7573e407362c1458a464a4d92fec2db4
ERR2136546_2.fastq.gz	63.9 MB	2026-10-31 15:16:12 EDT	a83a9dceaee29371219b98f0a2f1dc39

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ERR2136546_1.fastq.gz
ERR2136546_2.fastq.gz
```

Notes

- nearly as close: B-REF-BS4-40.fasta (Brucella suis bv. 4, 768/1000 hashes)
- MASH cannot separate Brucella canis and Brucella suis: ST 20 is Brucella canis in 70 of 70 isolate(s), which agrees with the closest reference

SRR19520329 ok Brucella melitensis ST 11

Identification: the MASH distances

Closest reference	NZ_CP007760-NZ_CP007761.fasta
What it is	Brucella melitensis bv. 3 str. Ether
NCBI accessions	NZ_CP007760.1, NZ_CP007761.1
MASH distance	0.003285
Estimated identity	99.671%
Shared min-hashes	875/1000 (at least 150 required)

Reference genome	Species	Biovar	Distance	Identity %	Shared hashes
NZ_CP007760-NZ_CP007761.fasta	Brucella melitensis	3	0.003285	99.671	875/1000
NC_012441-NC_012442.fasta	Brucella melitensis	2	0.003966	99.603	852/1000
NZ_CP018508-NZ_CP018509.fasta	Brucella melitensis	1	0.003996	99.600	851/1000
NC_003317-NC_003318.fasta	Brucella melitensis	1	0.004425	99.558	837/1000
NC_013119_NC_013118.fasta	Brucella microti		0.005411	99.459	806/1000
NC_006932-NC_006933.fasta	Brucella abortus	1	0.005477	99.452	804/1000
NZ_CP007682-NZ_CP007683.fasta	Brucella abortus	3	0.005510	99.449	803/1000
NC_022905-NC_022906.fasta	Brucella ceti		0.005808	99.419	794/1000
Bceti1Cudo.fasta	Brucella ceti		0.005976	99.402	789/1000
GCF_900446125.fasta	Brucella neotomae		0.006010	99.399	788/1000
NC_017251-NC_017250.fasta	Brucella suis	1	0.006078	99.392	786/1000
NC_010103-NC_010104.fasta	Brucella canis		0.006214	99.379	782/1000
B-REF-BS4-40.fasta	Brucella suis	4	0.006248	99.375	781/1000
NC_010169-NC_010167.fasta	Brucella suis	2	0.006420	99.358	776/1000
NZ_CP007719-NZ_CP007718.fasta	Brucella suis	3	0.006489	99.351	774/1000
NC_009505-NC_009504.fasta	Brucella ovis		0.007304	99.270	751/1000

The species call

Species reported	Brucella melitensis
Biovar	3 (of Ether, the closest reference of that species)
Decided by	MASH+ST
Species within reach	Brucella melitensis
Sequence type	11
What that ST is	ST 11 is Brucella melitensis in 546 of 546 isolate(s)

agree: The sequence type names the same species as the closest reference genome, which named it on its own: the two kinds of evidence agree.

MLST: the nine allele calls

Sequence type	11
Allele profile	3-2-3-2-1-5-3-2-10
Every locus called	yes
Reads in the input	1,136,924
Reads matching an allele	2,824

Locus	Allele	Reads	Runner-up	Its reads	Ahead	Ratio	Note
gap	3	350	gap_34	250	100	1.40	
aroA	2	324	aroA_1	256	68	1.27	
glk	3	282	glk_26	184	98	1.53	
dnaK	2	280	dnaK_6	204	76	1.37	
gyrB	1	374	gyrB_2	266	108	1.41	
trpE	5	348	trpE_31	242	106	1.44	
cobQ	3	260	cobQ_2	194	66	1.34	
int_hyp	2	328	int_hyp_20	222	106	1.48	
omp25	10	270	omp25_8	182	88	1.48	

Input files

File	Size	Modified	MD5
SRR19520329_1.fastq.gz	35.9 MB	2026-10-31 15:10:24 EDT	12edf200445bafd03d033fac3d98118c

File	Size	Modified	MD5
SRR19520329_2.fastq.gz	40.6 MB	2026-10-31 15:11:03 EDT	a81d3abf43d7a2849c9062061cd8603a

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SRR19520329_1.fastq.gz
SRR19520329_2.fastq.gz
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SRR8550484 ok Brucella suis ST 15

Identification: the MASH distances

Closest reference	NC_010169-NC_010167.fasta
What it is	Brucella suis bv. 2 str. ATCC 23445
NCBI accessions	NC_010169.1, NC_010167.1
MASH distance	0.024841
Estimated identity	97.516%
Shared min-hashes	422/1000 (at least 150 required)

Reference genome	Species	Biovar	Distance	Identity %	Shared hashes
NC_010169-NC_010167.fasta	Brucella suis	2	0.024841	97.516	422/1000
NC_017251-NC_017250.fasta	Brucella suis	1	0.025977	97.402	408/1000
NC_013119_NC_013118.fasta	Brucella microti		0.026060	97.394	407/1000
B-REF-BS4-40.fasta	Brucella suis	4	0.026143	97.386	406/1000
NC_010103-NC_010104.fasta	Brucella canis		0.026310	97.369	404/1000
NZ_CP007719-NZ_CP007718.fasta	Brucella suis	3	0.026310	97.369	404/1000
Bceti1Cudo.fasta	Brucella ceti		0.026648	97.335	400/1000
GCF_900446125.fasta	Brucella neotomae		0.027077	97.292	395/1000
NZ_CP018508-NZ_CP018509.fasta	Brucella melitensis	1	0.027164	97.284	394/1000
NC_022905-NC_022906.fasta	Brucella ceti		0.027250	97.275	393/1000
NZ_CP007760-NZ_CP007761.fasta	Brucella melitensis	3	0.027338	97.266	392/1000
NC_012441-NC_012442.fasta	Brucella melitensis	2	0.027601	97.240	389/1000
NZ_CP007682-NZ_CP007683.fasta	Brucella abortus	3	0.027778	97.222	387/1000
NC_003317-NC_003318.fasta	Brucella melitensis	1	0.027866	97.213	386/1000
NC_006932-NC_006933.fasta	Brucella abortus	1	0.027866	97.213	386/1000
NC_009505-NC_009504.fasta	Brucella ovis		0.028588	97.141	378/1000

The species call

Species reported	Brucella suis
Biovar	2 (of ATCC 23445, the closest reference of that species)
Decided by	MASH+ST
Species within reach	Brucella suis
Sequence type	15
What that ST is	ST 15 is Brucella suis in 99 of 103 isolate(s) (the rest: Brucella melitensis:2, Brucella ovis:1, Brucella sp.:1)

agree: The sequence type names the same species as the closest reference genome, which named it on its own: the two kinds of evidence agree.

MLST: the nine allele calls

Sequence type	15
Allele profile	1-2-7-1-3-3-5-3-2
Every locus called	yes
Reads in the input	1,693,672
Reads matching an allele	3,204

Locus	Allele	Reads	Runner-up	Its reads	Ahead	Ratio	Note
gap	1	508	gap_14	412	96	1.23	
aroA	2	284	aroA_16	232	52	1.22	
glk	7	282	glk_4	202	80	1.40	
dnaK	1	302	dnaK_27	224	78	1.35	
gyrB	3	384	gyrB_1	290	94	1.32	
trpE	3	346	trpE_5	296	50	1.17	
cobQ	5	294	cobQ_4	220	74	1.34	
int_hyp	3	444	int_hyp_1	272	172	1.63	
omp25	2	330	omp25_21	250	80	1.32	

Input files

File	Size	Modified	MD5
SRR8550484_1.fastq.gz	71.6 MB	2026-10-31 15:14:30 EDT	68833febebe273fadd9dc6ee0e95ed16
SRR8550484_2.fastq.gz	72.4 MB	2026-10-31 15:15:20 EDT	9b8438792c71722cd7fb4fb0d18e414c

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SRR8550484_1.fastq.gz
SRR8550484_2.fastq.gz
```

Definitions and methods

The species, from a MASH distance

The reads of a sample are reduced to the 1000 smallest hashes of their 21-mers, ignoring the k-mers seen once, and that sketch is compared with the sketch of each reference genome. The distance is an estimate of the fraction of positions that differ, so 0.000024 is about 99.998% identity. The closest reference names the species, provided the reads share at least the required number of its 1000 min-hashes; below that the sample is reported as ND, because the closest of sixteen genomes is not an answer.

Why the biovar is not a biovar call

The biovar in this report is the biovar of the closest reference genome of the species reported. The Brucella biovar references differ by very little, so the distance table in each sample block is the thing to read before quoting a biovar. For a biovar, use classical biotyping or an in silico Bruce-ladder.

The sequence type, from the alleles

The same reads are given to Seal with the alleles of the nine loci as references, counting each read for every allele it shares the highest number of 25-mers with and for no other. The allele the sample carries is therefore counted for every read of its locus, and a neighbouring allele only for the reads that fall where the two agree, so the allele with the most reads wins. No mismatch is allowed in a k-mer, because the alleles of a locus differ by as little as one base. Two alleles with the same count are reported as a tie rather than separated by a coin toss, and a locus below the minimum read count is not called.

What the ST values mean

A number is a profile that is in the table. "new" means all nine alleles were called but the combination is not in the table, which is either a genuinely new ST or a novel allele called as its nearest neighbour. "ND" means at least one locus had no call; the note says which.

How the species is decided when two are equally close

Some Brucella species are closer to each other than the variation within either of them: in the sketch that ships with BrucellaID, B. canis and B. suis bv. 4 are 0.00024 apart, the closest pair of the sixteen. Where a reference of a second species is within the close-match margin of the closest one, the distance has not really chosen, and the sequence type chooses among those species, because the two share no ST. An ST is only used when enough of its isolates agree on a species, and an ST that contradicts a species the distance picked cleanly is reported as a conflict and not obeyed.

What this report does not establish

That the sample is what the label on the tube says, that the library is not a mixture of two isolates, or that the reference data is current. It establishes what the software was given, what it measured, and what it concluded from those measurements.

Run information

Operator	M-O Duceppe
User account	[removed]
Computer	[removed]
Operating system	[removed]
Working directory	[removed]
Started	2026-10-31T21:21:05-04:00
Finished	2026-10-31T21:21:37-04:00
Command	python3 scripts/make_example_report.py reads

Software

Program	Version
brucellaid	1.0.0
Python	3.12.16
MASH	2.3
BBTools	40.05
Java	openjdk version "25.0.4.1" 2026-08-18

Parameters

Parameter	Value
Threads	8
Parallel samples	2
MLST	True
Input checksums	MD5

Reference data

Data	What it holds	Source	MD5
MASH sketch	16 reference genome(s), k = 21, 1000 hashes	the package's	44117b188407e97c5caff376f6e3b4eb
Reference table	16 genome(s) with their species, biovar, strain and accessions	the package's	d221a221f85377746fbbbb88906984dc
MLST alleles	309 allele(s) of the 9 loci	the package's	95fcf5e43f4cbf7ebe840c8e54bfbf46
MLST profiles	122 sequence type(s)	the package's	dd4bc06867bd58ce2a8f362755aa79a7
ST to species	118 sequence type(s) with a species	the package's	3a6751b8a1ab1273f3bc0e19e08efb7f

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.../brucellaid/data/references.msh
.../brucellaid/data/references.tsv
.../brucellaid/data/mlst_alleles.fasta
.../brucellaid/data/mlst_profiles.tsv
.../brucellaid/data/mlst_st_species.tsv
```

References

1. Whatmore AM, Perrett LL, MacMillan AP. Characterisation of the genetic diversity of Brucella by multilocus sequencing. BMC Microbiology 2007;7:34. doi:10.1186/1471-2180-7-34
2. Ondov BD, Treangen TJ, Melsted P, et al. Mash: fast genome and metagenome distance estimation using MinHash. Genome Biology 2016;17:132. doi:10.1186/s13059-016-0997-x
3. Jolley KA, Bray JE, Maiden MCJ. Open-access bacterial population genomics: BIGSdb software, the PubMLST.org website and their applications. Wellcome Open Research 2018;3:124. doi:10.12688/wellcomeopenres.14826.1
4. Bushnell B. BBTools. https://sourceforge.net/projects/bbmap/